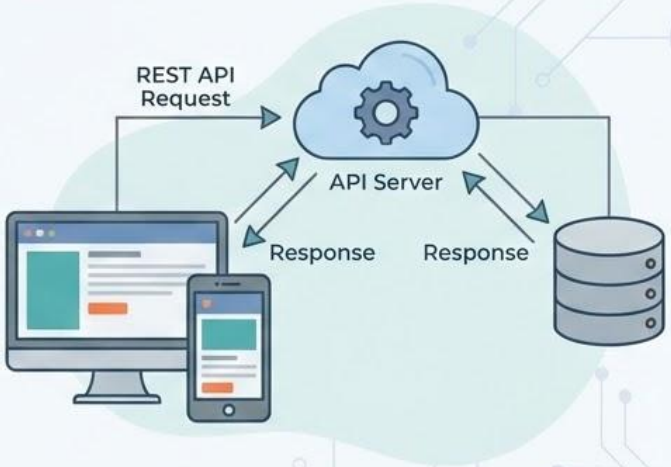
REST API Architecture

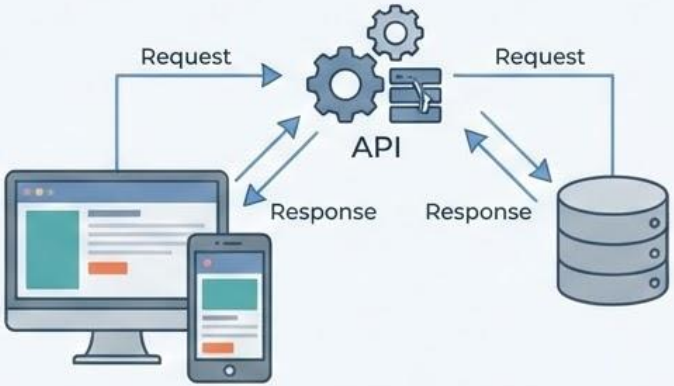
Design, Components, and Working of RESTful Web Services



Name : Sarthak Jagtap | Superset Id : 6746698 | College Name: Dr. D Y Patil Institute of Technology.

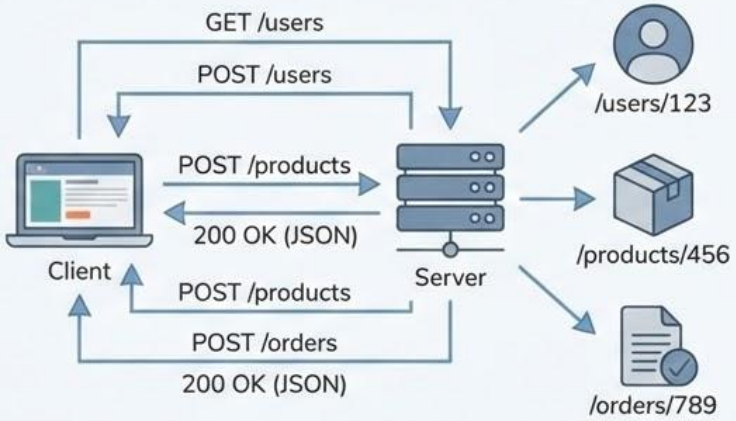
What is an API?

- API stands for Application Programming Interface
- It allows different software applications to communicate with each other
- Acts as an intermediary between client and server
- Exposes functionality without revealing internal implementation
- Promotes modular and reusable system design



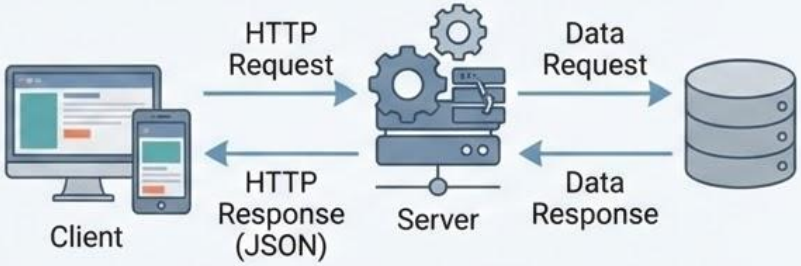
What is REST?

- REST stands for Representational State Transfer
- It is an architectural style for designing web services
- Uses standard HTTP protocol for communication
- Treats everything as a resource
- Communication between client and server is stateless



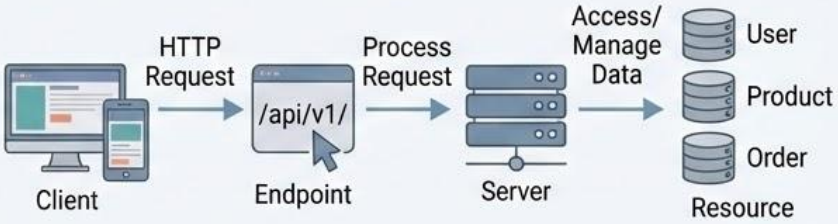
REST API Architecture Overview

- Client sends an HTTP request to the server
- Server processes the request
- Server returns an HTTP response
- Communication follows request–response model
- Data is commonly exchanged in JSON format



Core Components of REST API

- Client – Web or mobile application that sends requests
- Server – Backend system that processes requests
- Resource – Data entity such as user, product, or order
- Endpoint – URL used to access a resource
- HTTP Method – Defines the action to be performed on a resource

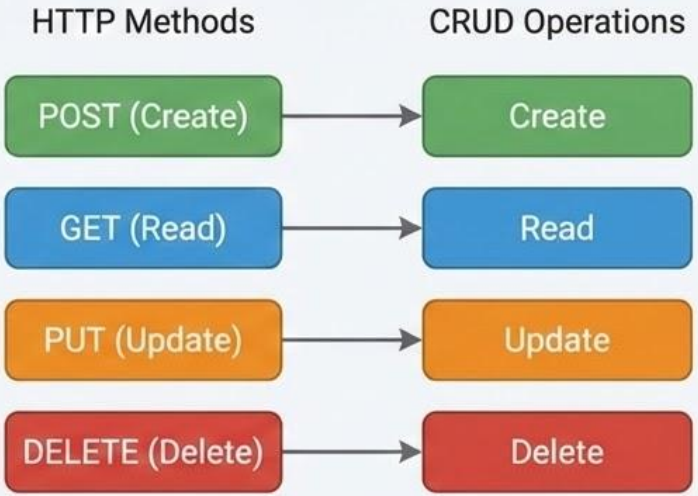


HTTP Methods in REST API

- GET – Retrieves data from the server
- POST – Creates a new resource
- PUT – Updates an existing resource
- DELETE – Removes a resource
- These methods correspond to CRUD operations

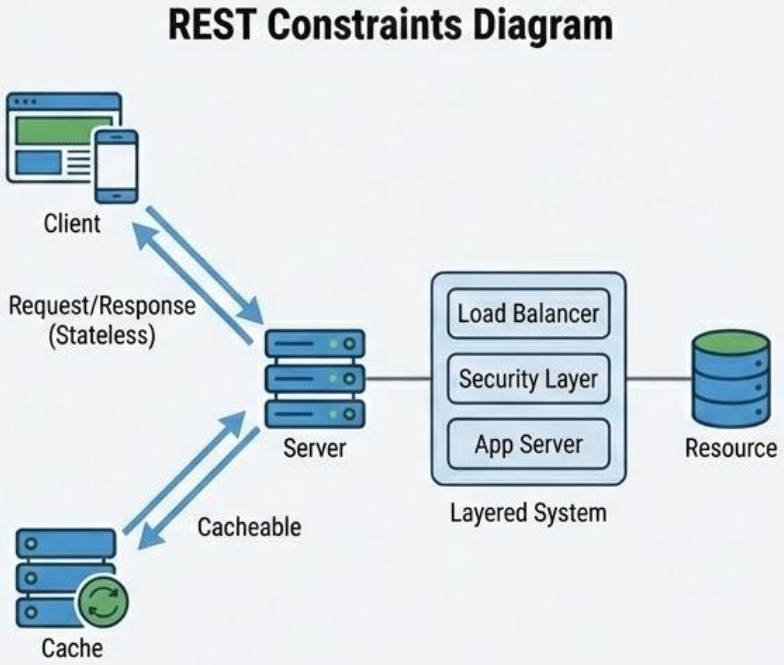
Examples:

- `GET /users`
- `POST /users`
- `PUT /users/1`
- `DELETE /users/1`



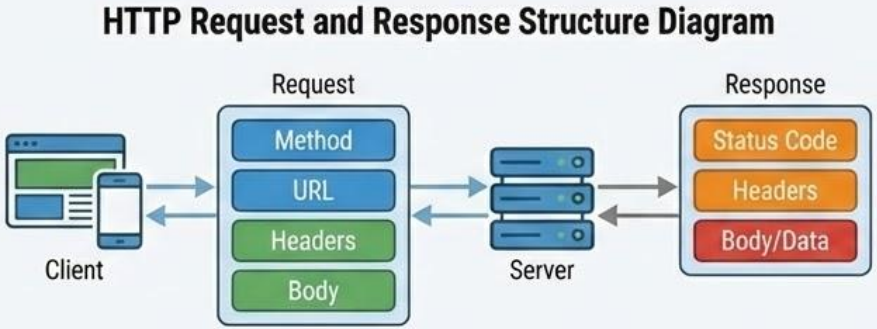
REST Architectural Principles

- Stateless communication between client and server
- Client–server separation of concerns
- Uniform interface for accessing resources
- Cacheable responses to improve performance
- Layered system architecture



Request and Response Format in REST API

- Request Includes:
- URL (Endpoint)
- HTTP Method
- Headers
- Body (optional)

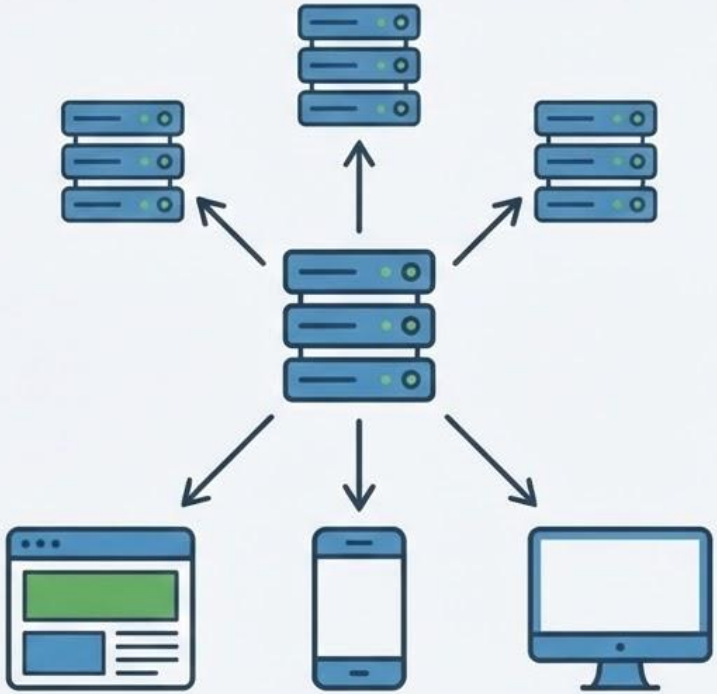


Response Includes:
- Status code
- Data (usually in JSON format)
- Message

Common HTTP Status Codes:
- 200 – OK
- 201 – Created
- 400 – Bad Request
- 404 – Not Found
- 500 – Internal Server Error

Advantages of REST API

- Simple and lightweight architecture
- Uses standard HTTP protocol
- Platform and language independent
- Highly scalable and flexible
- Easy integration with frontend applications
- Supports multiple client types (web, mobile, desktop)



Applications and Conclusion

Applications:
- Web applications
- Mobile applications
- Payment systems
- Cloud-based services
- Microservices architecture

Conclusion:
- REST API enables efficient communication between systems
- It follows simple, scalable, and stateless principles
- REST APIs are essential for modern software development